SARAVANAN K

M.Tech Biotechnology — Cell & Molecular Biology Researcher

saravananmtech25@gmail.com | +91 8072104819 | https://saravanank2503.github.io/saravana_k_portfolio/ | linkedin.com/in/saravanank25 |

Tiruchirappalli, Tamil Nadu

PROFESSIONAL SUMMARY

2025 M.Tech Biotechnology graduate with industry and academic research experience in molecular assay development, validation, and GLP-compliant documentation. Proven hands-on expertise in RT-PCR/qPCR, multi-biomarker ELISA, mammalian cell culture (BSL-2), and biomarker profiling. Supplementary computational fluency in RNA-Seq pipeline development.

WORK EXPERIENCE

Research Associate — Auxochromofours Solution Pvt Ltd, Bangalore *Sep 2024 – Present*

- Developed and validated multi-biomarker ELISA panels (12 markers), mtDNA copy number (qPCR), and telomere length (T/S ratio) assays with inter-assay reproducibility controls and GLP-compliant documentation
- Executed RT-PCR/qPCR workflows maintaining Ct CV <2%; designed custom primers using Primer3 and NCBI Primer-BLAST to reduce non-specific amplification
- Led heavy metal and trace element biomarker profiling across 18 analytes; managed external laboratory outsourcing and interpreted results for client decision-making
- Isolated high-integrity RNA/DNA from diverse biological matrices (TRIzol and column-based); performed NanoDrop (A260/280) and gel electrophoresis QC
- Authored 8 SOPs covering RT-PCR, ELISA, RNA extraction, and telomere assays; maintained full reagent traceability and calibration logs per GLP standards

Research Trainee — THSTI, Faridabad (Dr. Ruchi Tandon, HST Lab) *Dec 2024 – Apr 2025*

- Established NAFLD-CVD in vitro dual-cell model (HepG2 & H9c2) via FFA induction; identified 500 µM as peak hepatic steatosis threshold via Nile Red/DAPI fluorescence
- Detected CD36 and TNF-α upregulation by RT-qPCR (SYBR Green, ΔΔCq), confirming lipid uptake and inflammatory pathway activation; supported manuscript preparation
- Performed MTT cytotoxicity assays, compound screening across a focused bioactivity panel, TRIzol RNA isolation, cDNA synthesis; maintained BSL-2 cell culture with NanoDrop QC

Research Intern — NCSCM, Anna University & MS University

- Performed molecular docking (AutoDock Vina), virtual screening of marine alkaloids and phenolic compounds, and PyMOL structure visualisation

KEY PROJECT

Automated Reproducible RNA-Seq Analysis Pipeline *Personal Project | 2026*

- Built end-to-end containerized pipeline using Nextflow + Docker: FastQC → Trim Galore → STAR → featureCounts → DESeq2. Validated on CAR-T immunotherapy dataset (4 conditions × 3 replicates); generated volcano plots, PCA, and heatmaps. github.com/saravanank2503/Automated-RNA-Seq-Pipeline

EDUCATION

M.Tech Biotechnology (Integrated 6-Year) — Bharathidasan University, Tiruchirappalli *CGPA: 7.6 / 10.0|2019 – 2025*

CORE COMPETENCIES

- **Cell & Molecular Biology:** Mammalian cell culture (HepG2, H9c2), BSL-2 practices, RT-PCR/qPCR, RNA/DNA isolation (TRIzol & column-based), primer design (Primer3, NCBI Primer-BLAST), mtDNA copy number quantification, telomere length assay (T/S ratio), Nile Red & DAPI fluorescence staining, NanoDrop QC (A260/280), gel electrophoresis, cDNA synthesis, SDS-PAGE, Western blot.
- **Bioanalytical:** Multi-marker ELISA development & validation (12-marker panel), standard curve optimisation, cross-reactivity troubleshooting, MSP-qPCR (6 gene targets), MTT cytotoxicity assay, heavy metal & trace element profiling (18 analytes), GraphPad Prism, spectrophotometry.
- **Bioinformatics:** AutoDock Vina, PyMOL, SwissADME, Nextflow, Docker, RNA-Seq Analysis, NCBI BLAST, Primer3, R.
- **Quality & Compliance:** SOP authorship (8 SOPs), GLP documentation, assay validation (inter-assay & intra-assay), reagent traceability, equipment calibration logs, experimental design, outsourced laboratory management, result interpretation for client reporting.